ALGEBRA OF COMPLEX NUMBERS II

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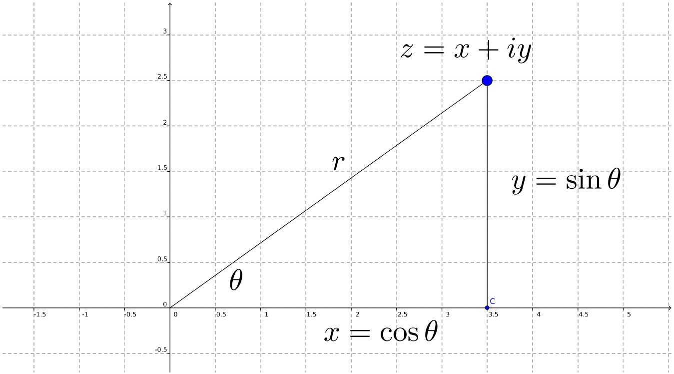
Lecture Outline



Definition (Trigonometric Form)

The trigonometric form of $z = x + iy$ is $z = r(\cos\theta + i\sin\theta)$ where $arg(z) = \theta$ and $\|z\| = r$.

Trigonometry and complex number

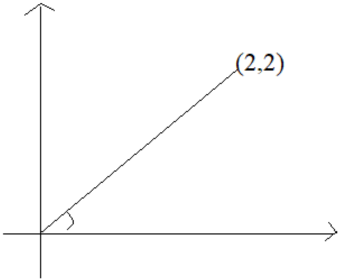
1. For a complex number $z = x + yi$ we can write the trigonometric representation $z = r(\cos\theta^* + i\sin\theta^*)$, where $r \in [0,\infty)$ and $\theta^* \in [0,2\pi)$ are the polar coordinates of the geometric image of z.
2. The polar argument θ^* of the geometric image of z is called the argument of z, denoted by $\arg z$.
3. The polar radius r of the geometric image of z is equal to the modulus of z.
4. The set $Arg z = \{\theta : \theta^* + 2k\pi, \quad k \in \mathbb{Z}\}$ is called the extended argument of the complex number z. Then

$$z = r(\cos\theta + i\sin\theta) = z = r\left[\cos\left(\theta^* + 2k\pi\right) + i\sin\left(\theta^* + 2k\pi\right)\right] \tag{1}$$

Example

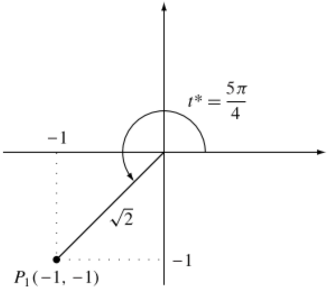
Find the trigonometric representation of the following numbers and determine their extended argument:

1. $z_1 = 2 + 2i$
2. $z_2 = -1 - i$
3. $z_3 = -1 + i\sqrt{3}$
4. $z_4 = 1 + i$



- $r_1 = \sqrt{2^2 + 2^2} = 2\sqrt{2}$
- $\theta_1^* = \tan^{-1}(1) = \dfrac{\pi}{4}$
- $z_1 = 2\sqrt{2}\left(\cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}\right)$
- $Arg\, z_1 = \left\{\dfrac{\pi}{4} + 2k\pi;\ k \in \mathbb{Z}\right\}$

$z_2 = -1 - i$

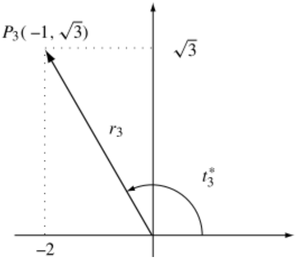


1. $r_2 = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$
2. Because is in the third quadrant

$$\begin{aligned}\theta_2^* &= \tan^{-1}\frac{y}{x} + \pi \\ &= \tan^{-1}(1) + \pi \\ &= \frac{\pi}{4} + \pi = \frac{5\pi}{4}\end{aligned}$$

3. $z_2 = \sqrt{2}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$
4. $Arg\, z_2 = \left\{\frac{5\pi}{4} + 2k\pi;\ k \in \mathbb{Z}\right\}$

$z_3 = -1 + i\sqrt{3}$



- $r_3 = \sqrt{(-1)^2 + \sqrt{3}^2} = 2$
- $\theta_3^* = \tan^{-1}(-\sqrt{3}) + \pi = \dfrac{\pi}{3} + \pi = \dfrac{2\pi}{3}$
- $z_3 = 2\left(\cos\dfrac{2\pi}{3} + i\sin\dfrac{2\pi}{3}\right)$
- $Arg\, z_3 = \left\{\dfrac{2\pi}{3} + 2k\pi;\ k \in \mathbb{Z}\right\}$

1 + i

$z = 1 + i$ has modulus $\sqrt{2}$ and argument $\arg(z) = \dfrac{\pi}{4} + 2k\pi, k \in \mathbb{Z}$.

In other words, the trigonometric form of z can be any one of the following

$$\begin{align}
z &= \sqrt{2}\,(\cos\pi/4 + i\sin\pi/4)\,; \quad k = 0 \tag{2}\\
&= \sqrt{2}\,(\cos(\pi/4 \pm 2\pi) + i\sin(\pi/4 \pm 2\pi))\,; \quad k = \pm 1 \tag{3}\\
&= \sqrt{2}\,(\cos(\pi/4 \pm 4\pi) + i\sin(\pi/4 \pm 4\pi))\,; \quad k = \pm 2 \tag{4}\\
&= \sqrt{2}\,(\cos(\pi/4 \pm 6\pi) + i\sin(\pi/4 \pm 6\pi))\,; \quad k = \pm 3 \tag{5}\\
&= \vdots \tag{6}
\end{align}$$

If there is no ambiguity, we choose the argument between 0 and 2π. Thus

$$z = \sqrt{2}\,(\cos\pi/4 + i\sin\pi/4) \tag{7}$$

Remarks

- $1 = \cos 0 + i \sin 0$
- $i = \cos \dfrac{\pi}{2} + i \sin \dfrac{\pi}{2}$
- $-1 = \cos \pi + i \sin \pi$
- $-i = \cos \dfrac{3\pi}{2} + i \sin \dfrac{3\pi}{2}$

If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ then

$$z_1 z_2 = r_1 r_2 \left[\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)\right]$$

If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2) \neq 0$ then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \left[\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)\right]$$

Example

Let $z_1 = 1 - i$ and $z_2 = \sqrt{3} + i$, then

$$z_1 = \sqrt{2}\left(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right), \quad z_2 = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

and

$$\begin{aligned} z_1 z_2 &= 2\sqrt{2}\left[\cos\left(\frac{7\pi}{4} + \frac{\pi}{6}\right) + i\sin\left(\frac{7\pi}{4} + \frac{\pi}{6}\right)\right] \\ &= 2\sqrt{2}\left(\cos\frac{23\pi}{12} + i\sin\frac{23\pi}{12}\right) \end{aligned}$$

Properties

Let z_1 and z_2 be two non-zero complex numbers. We have

1. $z_2 = z_1 \iff \|z_2\| = \|z_1\|$ and $\arg(z_2) = \arg(z_1) + 2k\pi, k \in \mathbb{Z}$.
2. $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) + 2k\pi$.
3. $\arg(z_1^n) = n\arg(z_1) + 2k\pi$.
4. $\arg(1/z_1) = -\arg(z_1) + 2k\pi$.
5. $\arg(\bar{z}_1) = -\arg(z_1) + 2k\pi$.

Euler and De Moivre's Form

Definition (Euler's equation)

The relation

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{8}$$

is called Euler's equation. Thus, if $z \in \mathbb{C} - \{0\}$

(Trigonometric form) $z = r(\cos\theta + i\sin\theta) \iff$ (Exponential form): $z = re^{i\theta}$.

Also, because any two arguments for a give complex number differ by an integer multiple of 2π we will sometimes write the exponential form as,

$$z = re^{i(\theta+2\pi k)}, \quad k = 0, \pm 1, \pm 2, \cdots \tag{9}$$

Forms

1. **Standard form** : $z = x + iy$
2. **Trigonometric form** : $z = r(\cos\theta + i\sin\theta)$ where $r = \|z\|$ and $\arg(z) = \theta + 2k\pi$.
3. **Exponential form**: $z = re^{i\theta}$ where $r = \|z\|$ and $\arg(z) = \theta + 2k\pi$.

Example

$z = 2 + 2i\sqrt{3}$ has modulus $\|z\| = 4$ and argument $\theta = \pi/3$.

Therefore,

$$z = 4e^{i\pi/3} \tag{10}$$

is its exponential form.

De Moivre's and Euler

De Moivre's theorem

For all real number θ and all integer n,

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta). \tag{11}$$

Definition (Euler's formula)

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} \qquad \cos(n\theta) = \frac{e^{in\theta}+e^{-in\theta}}{2} \tag{12}$$

$$\sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i} \qquad \sin(n\theta) = \frac{e^{in\theta}-e^{-in\theta}}{2i}. \tag{13}$$

Trigonometric identities

Let us use De Moivre's formula to express $\cos 2\theta$ in terms of $\cos\theta$ and $\sin\theta$

$$\cos 2\theta + i\sin 2\theta = (\cos\theta + i\sin\theta)^2 \quad \text{(by the theorem)} \tag{14}$$
$$= \cos^2\theta - \sin^2\theta + 2i\cos\theta\sin\theta \tag{15}$$

We observe that,

$$\cos 2\theta = \cos^2\theta - \sin^2\theta \tag{16}$$

and

$$\sin 2\theta = 2\cos\theta\sin\theta \tag{17}$$

Exercise

Express $\cos 3\theta$ *and* $\sin 4\theta$ *in terms of* $\cos\theta$ *and* $\sin\theta$.

The power of a complex number

De Moivre's Revisited

For $z = r(\cos\theta + i\sin\theta)$ and $n \in \mathbb{N}$, we have

$$z^n = r^n(\cos\theta + i\sin\theta)^n$$

$$z^n = r^n(\cos n\theta + i\sin n\theta)$$

Example

If $z = 2 + 2i\sqrt{3}$ find z^3

This has modulus $\|z\| = 4$ and argument $\theta = \pi/3$. so

$$z^n = r^n(\cos n\theta + i\sin n\theta) \tag{18}$$

$$z^3 = 4^3(\cos 3(\pi/3) + i\sin 3(\pi/3)) \tag{19}$$

$$z^3 = 64(\cos(\pi) + i\sin(\pi)) \tag{20}$$

$$z^3 = -64 \tag{21}$$

Thus

$$(2 + 2i\sqrt{3})^3 = -64$$

Example

Let us compute $(1+i)^{1000}$

The trigonometric representation of $1+i$ is $\sqrt{2}(\cos \pi/4 + i \sin \pi/4)$. Applying de Moivre's formula we obtain

$$\begin{aligned}
(1+i)^{1000} &= \sqrt{2}^{1000}(\cos 1000(\pi/4) + i \sin 1000(\pi/4)) \\
&= 2^{500}(\cos 250\pi + i \sin 250\pi) \\
&= 2^{500}
\end{aligned}$$

Roots of Complex Numbers

Definition (Complex polynomials)

Polynomials with complex coefficients and unknown are called **complex polynomials**.

$P(z) = iz^4 - 2z + i\sqrt{2} + 4$ is a complex polynomial of degree 4.

Theorem (Fundamental Theorem of Algebra)

Every complex polynomial equation of degree n has exactly n complex roots.

The roots of the complex polynomial $z^3 - 3z^2 + 2z$ are $0, 1, 2$. Note degree $= 3$ and number of roots $= 3$.

The n^{th} Roots of Unity

Consider a positive integer $n \geq 2$ and a complex number $z_0 \neq 0$. As in the field of real numbers, the equation

$$Z^n - z_0 = 0 \implies Z = \sqrt[n]{z_0} \tag{22}$$

is used for defining the n^{th} roots of number z_0.
Hence we call any solution Z of the equation (22) an n^{th} root of the complex number z_0.

Definition (Trigonometric)

Let $z_0 = r(\cos\theta + i\sin\theta)$ be a complex number with $r > 0$ and $\theta \in [0, 2\pi)$, then the number z_0 has n distinct n^{th} roots given by the formulas

$$Z_k = \sqrt[n]{r}\left(\cos\frac{\theta + 2k\pi}{n} + i\sin\frac{\theta + 2k\pi}{n}\right) \tag{23}$$

where $k = 0, 1, \cdots, n-1$

Alternatively definition for Exponential

If we had considered $z = re^{\theta i}$, the roots should have been

$$Z_k = \sqrt[n]{r}e^{\frac{\theta+2k\pi}{n}i} = \sqrt[n]{r}\exp\left(\frac{\theta + 2k\pi}{n}i\right); \qquad k = 0, 1, \cdots, n-1 \tag{24}$$

Example

Let us find the third roots of the number $z = 1 + i$

The trigonometric representation of $z = 1 + i$ is

$$z = \sqrt{2}\,(\cos \pi/4 + i \sin \pi/4)$$

The cube roots of the number z are

$$Z_k = \sqrt[3]{\sqrt{2}}\left[\cos\left(\frac{\pi}{12} + k\frac{2\pi}{3}\right) + i \sin\left(\frac{\pi}{12} + k\frac{2\pi}{3}\right)\right]; k = 0, 1, 2$$

or in explicit form as

$$Z_0 = \sqrt[6]{2}\left(\cos\frac{\pi}{12} + i \sin\frac{\pi}{12}\right)$$

$$Z_1 = \sqrt[6]{2}\left(\cos\frac{3\pi}{4} + i \sin\frac{3\pi}{4}\right)$$

$$Z_2 = \sqrt[6]{2}\left(\cos\frac{17\pi}{12} + i \sin\frac{17\pi}{12}\right)$$

Exercise

Find the argument and the standard form of the fourth roots of $a = \cos\dfrac{2\pi}{3} + i\sin\dfrac{2\pi}{3}$.

Solution

$$\arg(z) = \left\{\frac{\pi}{6}, \frac{4\pi}{6}, \frac{7\pi}{6}, \frac{10\pi}{6}\right\}$$

The n^{th} roots of unity

The roots of the equation $Z^n - 1 = 0$ are called the n^{th} roots of unity. Since $1 = \cos 0 + i \sin 0$, from the formulas for the n^{th} roots of a complex number (23) we derive that the n^{th} roots of unity are

$$Z_k = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}, \quad k = 0, 1, 2, \cdots, n-1 \tag{25}$$

Simplified as:

$$z_0 = 1,\; z_1 = e^{\frac{2}{n}\pi i},\; z_2 = e^{\frac{4}{n}\pi i},\; \ldots, z_{n-1} = e^{\frac{2(n-1)}{n}\pi i} \tag{26}$$

The n_{th} of unity in Exponential form

$$Z_k = e^{\frac{2k\pi}{n}i} = \exp\left(\frac{2k\pi}{n}i\right) \qquad k = 0, 1, 2, \cdots, n-1 \tag{27}$$

Example

If Z is a complex number, find the roots of $Z^3 = 1$

Solution

1. $Z_k = e^{\frac{2k\pi}{n}i} = \exp\left(\frac{2k\pi}{n}i\right) \qquad k = 0, 1, 2, \cdots, n-1$
2. $Z_0 = \exp\left(\frac{2(0)\pi}{3}i\right) = 1$
3. $Z_1 = \exp\left(\frac{2(1)\pi}{3}i\right) = \cos(2/3\pi) + i\sin(2/3\pi)$
4. $Z_2 = \exp\left(\frac{2(2)\pi}{3}i\right) = \cos(4/3\pi) + i\sin(4/3\pi)$

Complex Logarithm

Consider $z = re^{i(\theta+2k\pi)}$. By assuming that $\ln z$ exists, we use the properties of the logarithm function to simplify z.

We get $\ln z = \ln\left(re^{i(\theta+2k\pi)}\right) = \ln r + i(\theta + 2k\pi)$.

Definition

The **complex logarithm** of a complex number $z = re^{i\arg(z)}$ is defined as

$$\ln z = \ln r + i\theta \tag{28}$$

where θ is the argument of z that lies in the range $[-\pi,\ \pi]$.

In general, we use

$$\ln(z) = \ln r + i(\theta+2k\pi) \tag{29}$$

Example

$$\ln(-i) = \ln e^{-i\pi/2} \tag{30}$$
$$= \ln 1 - i\left(\frac{\pi}{2} + 2k\pi\right) \tag{31}$$
$$= -\frac{\pi}{2}i \tag{32}$$

Example

$$\ln\left(3e^{\frac{5\pi i}{3}}\right) = \ln 3 + i(\frac{5\pi}{3} + 2k\pi) \tag{33}$$
$$= \ln 3 + i\left(\frac{5\pi}{3} - 2\pi\right) \tag{34}$$
$$= \ln 3 - \frac{\pi}{3}i \tag{35}$$

Example

Find the trigonometric form of $z = 2^i$

$$\ln(z) = \ln\left(2^i\right) \tag{36}$$

$$= i\ln(2) = i\ln 2 \tag{37}$$

$$z = e^{ln(z)} \tag{38}$$

$$= e^{i\ln 2} \tag{39}$$

$$= \cos(\ln 2) + i\sin(\ln 2) \tag{40}$$

Example

Find the trigonometric form of $z = i^{2i+3}$

$$\ln(z) = (2i+3)\ln(i) \tag{41}$$

$$\ln(z) = (2i+3)\,i\left(\frac{\pi}{2}\right), \qquad \text{from(32)} \quad \ln(i) = i\left(\frac{\pi}{2}\right) \tag{42}$$

$$\ln(z) = -\pi + i\left(\frac{3\pi}{2}\right) \tag{43}$$

$$z = e^{\ln(z)} = \exp\left(-\pi + i\left(\frac{3\pi}{2}\right)\right) \tag{44}$$

$$z = e^{-\pi} e^{i\left(\frac{3\pi}{2}\right)} \tag{45}$$

$$z = e^{-\pi}\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right). \tag{46}$$

Alternative Solution

$$
\begin{aligned}
z &= i^{2i+3} && (47)\\
&= \left(e^{\frac{\pi}{2}i}\right)^{2i+3}, \qquad \text{NB} \quad i = e^{\frac{\pi}{2}i} = \cos(90) + i\sin(90) && (48)\\
&= e^{-2\frac{\pi}{2}+\frac{3\pi}{2}i} && (49)\\
&= e^{-\pi}e^{\frac{3\pi}{2}i} && (50)\\
&= e^{-\pi}\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right). && (51)
\end{aligned}
$$

Exercise

1. *Find the square and cubic roots of the following complex numbers:*
    1. $z = 1 + i$
    2. $z = i$
    3. $z = 1/\sqrt{2} + i/\sqrt{2}$
2. *Find the fourth roots of the following complex numbers:*
    1. $z = -2i$
    2. $z = \sqrt{3} + i$
    3. $z = -7 + 24i$
3. *Solve the equations:*
    1. $z^3 - 125 = 0$
    2. $z^4 + 16 = 0$
    3. $z^7 - 2iz^4 - iz^3 - 2 = 0;$

Exercise

1. *Find the polar coordinates for the following points, given their cartesian coordinates*
   1. $M_1 = (-3, 3)$;
   2. $M_2 = (-4\sqrt{3}, -4)$;
   3. $M_3 = (0, -5)$;
2. *Find the cartesian coordinates for the following points, given their polar coordinates:*
   1. $P_1 = (2, \pi/3)$
   2. $P_2 = (3, -\pi)$
   3. $P_3 = (1, \pi/2)$
3. *Find polar representations for the following complex numbers*
   1. $z_1 = 6 + 6i\sqrt{3}$
   2. $z_2 = -4i$
   3. $z_3 = \cos a - i \sin a, \quad a \in [0, 2\pi)$

END OF LECTURE

THANK YOU